

HAT-P-12b

R-0673

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_260410 · created 2026-07-27T18:49:03+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461140.9056 +/- 0.0052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.187 +/- 0.053
Transit depth (Rp/Rs)^2	3.51 +/- 1.98 [%]
Orbital Inclination (inc)	88.22 +/- 1.69
Airmass coefficient 1 (a1)	1.173 +/- 0.056
Airmass coefficient 2 (a2)	-0.132 +/- 0.096
Scatter in the residuals of the lightcurve fit is	4.84 %
Best Comparison Star	#2 - [366, 102]
Optimal Aperture	3.21
Optimal Annulus	10.5
Transit Duration (day)	0.085 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.187 ± 0.053 vs 0.1406 (deviation 0.88σ)
Duration fit vs published	0.085 d vs 0.0975 d (deviation 0.89σ)
Mid-time O–C	-10.9 min
Depth SNR	3.5
Residual scatter	4.84 %

Light curve

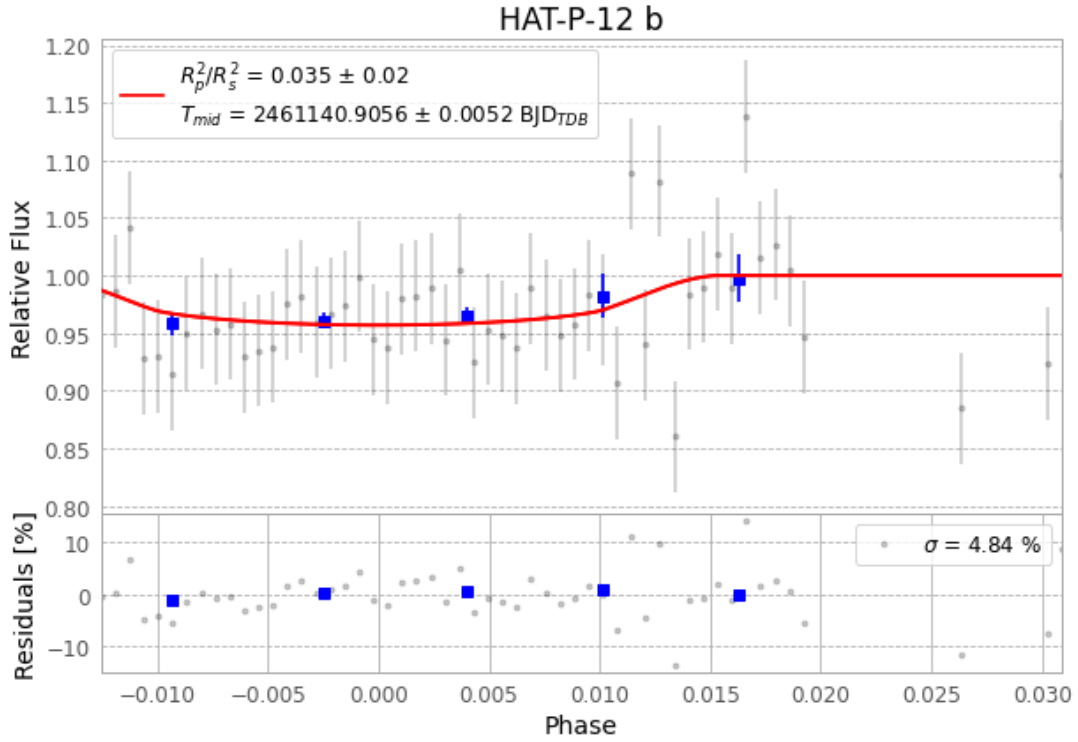


Figure 1 — FinalLightCurve_HAT-P-12 b_2026-04-10.png (SHA-256 8747f47d72b50e71...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [297, 213], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5297ae6b42bbf4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

68 FITS frames (45 MB), HATP-12260410083916.FITS → HATP-12260410120016.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-260410.log (42c30d0d6fec...), FinalLightCurve_HAT-P-12 b_2026-04-10.png (8747f47d72b5...), FinalLightCurve_HAT-P-12 b_2026-04-10.csv (b2f867468380...)

Compute resources

Wall time 295.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 18:49Z.